

INTERNSHIP PRESENTATION

Project Title: *Loan Default Prediction using Neural Networks*

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PROBLEM MOTIVATION

- Loan defaults lead to significant financial losses
- Need for reliable risk prediction systems
- Real-world data is highly imbalanced
- Accuracy alone is not sufficient for evaluation

PROBLEM STATEMENT

Objective: Predict whether a loan applicant will default

Problem Type: Supervised binary classification

Input Features:

- Financial (income, credit amount, annuity)
- Demographic (age, employment history)

Output Classes:

- 0 → Non-default
- 1 → Default

Key Challenge:

- Highly imbalanced dataset (~92% non-default, ~8% default)
- Standard models biased toward majority class

DATASET & PREPROCESSING

Dataset Overview

- ~307,000 records
- 120+ features

Feature Types:

- Numerical (income, credit, annuity)
- Categorical (gender, occupation, contract type)

Preprocessing Steps:

- Missing value handling (median / most frequent)
- One-hot encoding for categorical variables
- Feature scaling (standardization)

Data Split:

- Train–Test split with preserved class distribution

BASELINE MODEL

Model Type: Multilayer Perceptron (MLP)

Architecture:

- Input Layer
- Hidden Layers: 128 $\rightarrow$ 64 neurons
- Output Layer: Sigmoid

Configuration:

- Activation: ReLU
- Optimizer: Adam
- Loss: Binary Cross-Entropy

Initial Performance:

- Accuracy $\approx$ 0.92
- Recall $\approx$ 0.01 (very low)

PROBLEM WITH BASELINE MODEL

- High accuracy but poor real-world performance
- Very low recall for defaulters
- Fails to identify high-risk applicants
- Accuracy is misleading for imbalanced data
- Model biased toward majority class (non-default)

OPTIMIZATION METHODS

- **Class Weighting:** Handle class imbalance
- **Dropout (0.3):** Reduce overfitting
- **Learning Rate Scheduling:** Improve convergence
- **Early Stopping:** Prevent overfitting
- **Threshold Tuning:** Balance precision & recall
- **Batch Normalization:** Stabilize training

RESULTS COMPARISON

Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Baseline	0.92	0.55	0.01	0.02
Optimized	0.83	0.22	0.45	0.30

- F1 Score increased
- Accuracy decreased due to better minority detection
- Recall improved significantly

THRESHOLD ANALYSIS

Effect of Threshold on Performance

Threshold	Precision	Recall	F1 Score
0.3	0.11	0.87	0.20
0.4	0.13	0.76	0.23
0.5	0.16	0.64	0.26
0.6	0.18	0.52	0.27 (Best)

- Lower threshold → High recall, low precision
- Higher threshold → Better precision, lower recall
- Selected threshold: 0.6 (best balance)

KEY LEARNINGS

- Accuracy is not reliable for imbalanced datasets
- Class imbalance must be handled explicitly
- Recall is critical for risk prediction problems
- Threshold tuning improves real-world performance
- Model evaluation should use multiple metrics

CONCLUSION

- Optimization improved recall significantly
- Achieved better balance (F1 score improved)
- Model is more suitable for real-world risk prediction
- Baseline model failed to detect defaulters

Thank

You